

EDUCATION

Worcester Polytechnic Institute	May 2025
B. S. Robotics Engineering; B. S. Computer Science	Worcester, MA
◦ Relevant Classes: <i>Deep Learning For Perception, Swarm Intelligence, AI for Robotics, Unified Robotics: Manipulation, Navigation</i>	

DOMAIN SKILLS

Perception & Sensing	Intel Realsense, LiDAR, Image Segmentation, Object Detection, Kalman Filters, YOLOv11
Motion Planning	ORCA, SLAM, (gmapping), AMCL, A*, Pure Pursuit, Monocular Depth Estimation
Simulation	Nvidia IsaacSim, Gazebo, Unity Game Engine, MuJoCo, Blender (Synthetic Data Generation), PyGame
ML / RL Tools	PyTorch, Scikit-Learn, Nvidia IsaacLab, DQN, Imitation Learning, ResNet, Pandas, W&B
Embedded Systems	MicroPython, Arduino, Raspberry Pi, Nvidia Jetson, UART/I ² C, ROS/ROS2, HALs
Version Control & Dev Tools	Git, Agile, Kanban, Azure DevOps, Jira, Figma, Github Projects, Microsoft Office
Languages	Python, TypeScript, C++, C#, Java, C, JavaScript, Kotlin, MATLAB

WORK EXPERIENCE

Smartapp.com – Robotics Branch	May 2024 – Aug 2024
<i>Autonomy Engineering Intern</i>	Worcester, MA
▪ Expanded on a Python and React-TS thrust test stand hosted on a Raspberry Pi, with customizable datalogging and test procedures	
▪ Developed an intuitive and flexible motor library, which enables ‘hot-swapping’ of intelligent motor classes and objects	
▪ Optimized proprietary robot locomotion techniques in Nvidia’s IsaacSim using deep reinforcement learning (PyTorch and PPO)	
OpenSTEM: Experiential Robotics Platform (XRP)	Aug 2022 – Present
<i>Lead Software Developer, XRPLib</i>	Worcester, MA
▪ Designed and maintained a modular MicroPython-based robotics control library deployed across 20K+ classroom-scale robots	
▪ Managed a team of 1-5 over 2.5 years while coordinating with corporate partners and engineers from DEKA and Sparkfun	
Jacobs Technology – Jacobs Software Engineering Center	May 2022 – Aug 2022
<i>Software Engineering Intern</i>	Hudson, NH
▪ Refactored and extended C#/.NET mission support tooling to manage dynamic SQL filter sets for DAFIF airspace pathing datasets	

PROJECTS

Terrawarden Drone Cleanup – Major Qualifying Project	Aug 2024 – May 2025
▪ Developed aerial manipulator for autonomous waste collection using Intel RealSense and YOLOv11 with low-latency inference (<3ms)	
▪ Implemented object detection model trained on procedurally rendered Blender dataset, enabling robust generalization to roadside litter	
▪ Architected ROS2-based state machine to isolate high-level behavior, programmed using Test-Driven Development (PyTest)	
Drone Racing Gate Semantic Segmentation	Aug 2024 – Oct 2024
▪ Used Blender to generate a dataset of 5000 images of drone racing gates for training a semantic segmentation model (U-Net)	
▪ Applied various image augmentation techniques, such as gaussian blur and color jitter, to increase the robustness of the trained model	
Gesture-Controlled Claw Machine	Jan 2025 - Jan 2025
▪ Designed, assembled, and programmed a hand-gesture controlled claw machine using spare 3D printer parts for GoatHACKs 2025	
▪ Synced claw machine motion to user hand movements using an RGB camera and Google’s Hand Landmarker model on a Raspberry Pi	
Video Game AIs	Jan 2024 – Mar 2025
▪ Benchmarked the abilities of an algorithmic model (Expectiminimax) vs a reinforcement model (Q-Learning) at playing Bomberman	
▪ Implemented imitation learning on a Deep Q-Learning model in Pytorch, training it to play the Snake Game	
Robotic Navigation – SLAM and AMCL	Oct 2023 – Dec 2023
▪ Developed a robot that could autonomously navigate and map an unknown space, then localize itself when relocated at a later time	
▪ Implemented Simultaneous Localization and Mapping (SLAM), AMCL, A*, and Pure Pursuit on a Turtlebot3 with a planar LiDAR	
Hospital Wayfinding & Facilities System	Mar 2023 – May 2023
▪ Developed JavaFX-based kiosk application with Dijkstra-based pathfinding, signage customization, and service requests over SQL backend	
▪ Led 11-member Agile team in design, implementation, and user testing phases with modular MVC and OOP design patterns.	

EXTRACURRICULARS

WPI Robotics Prototyping Club, Founder, Treasurer	Aug 2024 - Present
WPI Cooking Club, President	Apr 2023 - Present
WPI Rho Beta Epsilon, Alpha Chapter	Feb 2025 - Present
WPI Bowling Club, Treasurer	Aug 2022 – Feb 2023
WPI VexU, Software Co-Lead	Aug 2022 – Feb 2023
Scouting America, Troop 106, Eagle Scout	Mar 2014 – July 2021